

Abstract

This paper reviews the literature regarding the effect that succeeding in high school mathematics has on college success. Researchers have established that success in high school mathematics is strongly correlated with success in college and higher earnings. More recent research seeks to show that success in high school mathematics actually *causes* success in college. Such analyses are difficult because researchers must draw meaning from naturally occurring data, rather than through experimentation. Researchers have employed many different econometric methods with varying levels of success. Studies using the best methods suggest that success in high school mathematics does in fact cause later educational success, but more studies should be done to confirm these findings. The paper ends by discussing areas for further research.

Keywords: curriculum effects, mathematics education, education policy, education reform, education policy, mathematics, high school mathematics, algebra, geometry, college success indicators, econometrics, economics of education

The Effect of Success in High School Mathematics on College Success

Education is paramount to the success of society. Individuals in a more educated society can understand and take care of each other. An educated individual provides for their own needs and leads a more fulfilling life. This literature review explores the effect that high school mathematics education has on college success. Knowing if mathematics has a positive impact on overall academic success is important because if it does, both government and privately funded educational institutions should raise math teacher salaries, improve teacher training, and fund more studies about how to teach math well.

For years, mathematics has been one of the core subjects taught in school. Success in Mathematics is often viewed as being critical to overall success in school. Does succeeding in high school mathematics cause college success? Many studies have shown a strong correlation between success in mathematics and overall educational success (Altonji, 1992; Gamoran & Hannigan, 2000; Levine & Zimmerman, 1995; Smith 1996). However, these results do not imply that success in mathematics causes overall educational success. This question is difficult to answer, because the data related to the subject is affected by many confounding variables. To account for these confounding variables, some researchers use advanced statistical methods such as propensity score matching, instrumental variables, and regression discontinuity design. This paper takes a deeper look into why it is so difficult to discover if success in high school mathematics causes college success and examines the methods that have been used to overcome these difficulties. Certainly, success in high school mathematics is correlated with college success, and current research is beginning to suggest that, indeed, succeeding in high school mathematics causes college success. Researchers should focus on utilizing sound statistical methods in future studies to confirm these findings.

The Inherent Problems

Even Altonji (1992), who largely pioneered research in the effects of curriculum design on college and career success, recognized the multitude of problems that come with seeking to answer this question. The easiest way for researchers to establish causation would be to do a simple experiment: randomly assign students to take different math classes. In order to control the experiment more accurately, researchers could mandate that the students take all the exact same classes except for their math classes. Clearly, experimenting on people's lives this way is unethical. Thus, researchers must extract meaning from existing data about student performance. Seeking to extract meaning from existing data leads to many difficulties in establishing causation. All of the literature that seeks to show that success in high school math causes college success touch on some of these difficulties (Allensworth, Nomi, Montgomery, & Lee, 2009; Altonji, 1992; Byun, Irvin, & Bell 2015; Dougherty, Goodman, Hill, Litke, & Page, 2017; Joensen & Neilsen, 2009; Kim, Kim, DesJardins, and McCall 2015; Levine & Zimmerman, 1995; Rose & Betts, 2001). However, the most thorough expositions are given by Altonji (1992) and Allensworth et al. (2009). They point out selection bias on an individual level, selection bias on a school level, effects of socioeconomic standing, effects of parents, and differences in instructor quality.

As an example of selection bias in this context, consider that the Virginia department of Education (2001) showed that students who perform well in Algebra II in high school perform better in college. We cannot use this fact to argue that the math class caused them to perform better in college because the sample of students who took Algebra II is not random: more advanced students will naturally be more likely to take more advanced classes. Furthermore, even using students' prior test scores to control for the lack of random selection can be

dangerous, as many skills that help students succeed go unmeasured, especially before high school (Altonji, 1992). As Altonji (1992) noted, parents who push their children to attend college are also likely to pressure their children to take higher levels of math in high school. This student level selection bias plagues the current study.

Byun et al. (2015) declare that assuming advanced math courses have the same benefits for students of different socioeconomic backgrounds is dangerous. Thus, researchers should control for student backgrounds in their analyses. Though disregarding students' socioeconomic backgrounds will still give information about the effects of the courses, researchers will gain more information and more accurate information by considering it as part of the analysis.

Now, suppose that to avoid student level selection bias researchers compare students across schools. For example, they could compare college success for students who attended a school that requires Algebra II to college success for students who attended a school that does not require Algebra II. Comparing across schools still fails to solve the problem, because the schools are not randomly selected. Allensworth, et al. (2009) pointed out that most likely, the school that requires Algebra II also has a stronger culture of preparing students for college. This method introduces a school level selection bias.

Altonji (1992) brings up the point that differences in difficulty or quality of instruction in a course can have unseen effects on a student's later success. These disparities create bias across schools and even across teachers at the same school. Additionally, standards for what a student must understand to pass a given class vary across states, schools, teachers, and, as Schneider (2009) pointed out, across years. Thus, it is difficult to measure the effects of learning advanced math because it is difficult to measure what a student has actually learned in comparison to another student. Dougherty et al. (2017) and Kim et al. (2015) control for the issue by studying

subjects who can be compared to one another by some standardized test which all students took. In this case researchers must recognize that even standardized tests give an imperfect representation of student knowledge.

Methods Used in Overcoming the Problem

Clearly, many factors confound the data about the effects of mathematics education on college success. There are enough confounding factors that enumerating them all is impossible. Thus, it is critical to use statistical methods that will control for the seen and unseen factors affecting the data. Because controlling for all factors manually is problematic, the best way to obtain a clean experiment is to analyze the data in a way that gives a sense of randomness to how students were assigned to take certain courses.

Natural Experiments

As discussed above, standard statistical methods cannot be used because of the problem of comparison. Researchers can look at what happened when a given student takes high level math classes in high school but cannot know what would have happened if he or she had not taken those classes. They also cannot randomly assign students to take certain classes. A natural experiment is when a naturally occurring circumstance gives a sense of random assignment. If a natural experiment supplies complete randomness researchers do not need to worry about confounding variables. However, most natural experiments provide partial but not complete randomness. When using results from a natural experiment, the researcher must be aware of which confounding variables the natural experiment did not control for and use further statistical methods to account for them.

As an example of a natural experiment, Allensworth et al. (2009) study a situation where, in 1997, Algebra I and English I were suddenly required for all ninth graders in Chicago. Thus,

they could study the effect that being mandated into a higher math class had on the students, using the group of students from the year before the policy change as the control group. Taken as a whole, assigning students to take Algebra I one year but not the year before could be considered mostly random. However, it is still important to be aware of other circumstances. For example, researchers must also control for prior test scores because the group mandated to take Algebra I could have already been performing better on standardized tests than the students before them.

Propensity Score Matching

Propensity score matching (PSM) seeks to obtain a sense of random assignment by matching each experimental subject with another experimental subject who was not given treatment. For example, suppose researchers want to study the effects of taking Algebra II in high school. Using the method of PSM, to see the effect of having taken Algebra II, they would match each student who took Algebra II with a student who did not take Algebra II of similar socioeconomic standing, parental attitudes, past performance, and all other possible characteristics that could affect the study. Though PSM can be used to control for measurable variables, methods that show random assignment are superior because they control for all variables—both measured and unmeasured.

Byun et al. (2015) utilized PSM to analyze data from the Educational Longitudinal Study of 2002-2006. They matched students on a wide array of characteristics, including socioeconomic status, prior academic achievement, and parental educational expectation. They found the positive impact that taking advanced math classes has on college enrollment can mostly be explained by other factors such as family and school influences. They also found that taking advanced math classes has less of a positive impact for Black students than for White

students. Their methods are fairly sound. However, as previously mentioned, since PSM relies on matching based on observable characteristics, there is always some level of inaccuracy due to unobservable or unseen characteristics.

Instrumental Variables

Instrumental variable estimation is another way of seeking to establish a causal relationship between two variables when a classical experiment is not feasible. In order for an instrumental variable estimation to establish causation, there must be an instrumental variable that is correlated with the independent variable but not correlated with the dependent variable. As an example of a case where instrumental variable estimation is useful, Murray (2006) discussed the effect of incarceration rates on crime. In this case, measurements related to prison overcrowding lawsuits are valid instruments because they are correlated with incarceration (less people were incarcerated during overcrowding lawsuits), but not with crime rates. In the case examined by this paper, the independent variable is how well a student learns mathematics in high school and the dependent variable is how well the student performs in college. Therefore, a valid instrument must be correlated with mathematical learning in high school but not with college success.

Kim et al. (2015) analyzed data regarding students in Florida, following them from grades 7-12 and into college and the workplace. They used county unemployment levels during the ninth grade year of the subjects of interest as their instrumental variable. They concluded that completing Algebra II in high school has a significant impact on a student deciding to attend college, but no impact on their completion of college. However, their instrumental variable fails to meet the criteria for being a valid instrument. It seems unlikely that unemployment during a student's ninth grade year has no correlation with their decision to attend and perform well in

college. They support their claim by showing that the unemployment rate changes enough from year to year that it won't have an impact. However, if unemployment has an impact on what math classes a student chooses to take in ninth grade, as they claim it does, it would be likely that it also impacts many other variables that would affect student's decision to attend and perform well in college.

Altonji (1992) introduced the idea of using variation of curriculum across high schools (specifically, number of courses taken in each subject) as an instrumental variable, and found that taking one additional academic course in high school yields little effect on future success, but the effect of taking additional math is greater than the effect of taking more of any other subject. His methods rely on two assumptions: that schools offer identical courses to students but require different courses than one another and that the differences in course requirements are unrelated to student characteristics and any unobserved characteristics (Altonji 1995). By his own admission, these assumptions are not realistic. Rose and Betts (2001) use the same method, but apply it more specifically to mathematics. They also improve on his method by adding more controls for school effects.

More recently, Joensen and Nielsen (2009) have built upon the approach of Altonji (1992) and Rose and Betts (2001) by using instrumental variables. However, the natural experiment that they took advantage of led to a stronger instrumental variable that can be trusted with more certainty. They studied high schools in Denmark. In the 1980s, there was a pilot program for a new curriculum instituted at some schools—a curriculum giving students more flexibility in their course options. In the pilot program, some students took more advanced math than they would have without it. This natural experiment does have some bias, as the schools self-selected to be a part of the pilot program and students still selected to take math. As their

instrumental variable, Joensen and Neilson (2009) use whether or not a student's high school decided to enroll in a pilot program *after* the student was enrolled in the high school. They add additional controls for the preexisting qualifications of students. They find that those who take more mathematics than they would have without the pilot curriculum received a higher education and have higher earnings later on.

Regression Discontinuity Design

Regression discontinuity design (RDD) is another way of extracting randomness from a situation that does not seem random. RDD is achieved by comparing only subjects close to some arbitrary cutoff to obtain a sense of randomness.

Dougherty et al. (2017) use regression discontinuity design for their analysis. They observed the results of a pilot program in North Carolina that placed many students on a higher math track than normal based on a placement test. Obviously, assigning students to take a different math class based on a test is not random assignment. However, it could be considered random assignment for those students who scored close to the cutoff on the test. Suppose that students who scored 80% or higher on the placement test were placed into the higher-level math class. It is likely that the students who scored 79% and the students who scored 80%—or even 81%—have nearly identical mathematical abilities. Thus, by only comparing students who scored close to the cutoff, they obtain a sense of randomness. They conclude that accelerated math placement has a positive effect on college readiness. These results are near certain, as Dougherty et al. (2017) used sound methods for establishing random assignment.

Discussion

Of the surveyed literature that seeks to show causation, Allensworth et al. (2009), Byun et al. (2015), and Kim et al. (2015) claim no effect, while Altonji (1992), Levine and

Zimmerman (1995) Rose and Betts (2001), Joensen and Neilsen (2009), and Dougherty et al. (2017), claim that succeeding in high school mathematics leads to improved college performance. In general, the studies that found that success in high school mathematics causes college success used sounder statistical methods (Altonji, 1992; Levine & Zimmerman, 1995; Rose & Betts, 2001; Joensen & Neilsen, 2009; Dougherty et al., 2017). However, all of the literature this paper has reviewed, without exception, has based their statistical methods on imperfect assumptions. Examining the best research papers, those by Joensen and Nielsen (2009) and Dougherty et al. (2017), reveals a number of ideas that future researchers should learn from: research should be done in collaboration between econometricians and education researchers, take advantage of natural experiments to establish randomness, and use statistical methods grounded on believable assumptions to establish causation.

Research in the area of curriculum effects should almost certainly be done in collaboration between econometricians and education researchers. Experts in education know the system they are studying and are passionate about the topic. For example, Allensworth et al. (2009) utilize a complex model about the structure of the education system to help them in their study. However, education researchers often lack the data analysis skills that econometricians use to extract causation from naturally occurring data. Not only do econometricians and education researchers need to collaborate on this topic, they need to read each other's research. Allensworth et al. (2009), the Virginia Department of Education (2001), and Douglas and Attewell (2017) give no reference to and appear to have not even read Altonji's (1992) study, which is considered foundational by econometricians and expounds upon many of the difficulties in this area of research (Joensen and Nielson, 2009; Levine and Zimmerman, 1995; Rose and Betts 2001).

Researchers also need to make an earnest effort to establish causation. Studies such as those done by Douglas and Attewell (2017) and the Virginia Department of Education (2001) reiterate the correlation between high school math success and college success—but this has already been established. Future research should seek a case for causation by finding natural experiments and using sound methods to analyze the resulting data.

In his book, *The Math Myth*, Andrew Hacker (2016), a Political Scientist, argues that the only reason success in mathematics correlates so strongly with college success is that the current educational system mandates math for college entrance exams and general education requirements. Although he does not have evidence to support his claim, he does pose an interesting question that is worth investigating: Would success in high school mathematics lead to college success even if it was not required for entrance exams and general education? There is no scholarly research, as of right now, that tries to answer this question. It is certainly a question that education researchers and econometricians should study.

Dougherty et al. (2017), Smith (1996), and Gamoran and Hannigan (2000) found that earlier exposure to Algebra I for students who would have usually been placed in remedial courses was positively correlated with students' college preparation. These results beg the questions: What were the negative effects on the students who were pushed to Algebra I and then failed out? Were they even less prepared for college than they would have been otherwise? These studies mention but do not focus on or extrapolate upon these questions, but they are important questions worthy of more attention from researchers.

Conclusion

It is important to understand the impact that mathematics education has on overall educational success so that local, state, and national leaders can make better policy and

curriculum decisions. The strong correlation between success in high school mathematics and college success has been known for some time. More recent studies done with various statistical methods suggest that learning math well in high school actually causes success in college. Researchers should do more studies to confirm these findings, using natural experiments where a sense of random assignment can be used to establish causation. There should be more collaboration between education researchers and economists doing research in education, as they could both benefit from one another's perspectives. Further research should also be done on related questions: Does pushing students into Algebra I sooner have a negative effect on those who fail out? Would a strong high school math background increase college success even without requiring it on entrance examinations? These questions should drive future research and policy decisions.

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